

Tutorial 8 - SPM (wk13 - done)

Question 1

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Question 1

Describe TWO (2) examples of critical systems, which include a *mission-critical* system and a *business-critical* system. For each system that you have given, assess which dimension of system dependability is the most important. Provide justifications in your answer.

For revision***

at some point in time the system **does not** deliver a service as expected by its users

8.1 Critical Systems

There are 3 types of **critical systems**:

a. Safety-Critical systems

- A **system failure** may cause:
 - injury,
 - loss of life or
 - major environmental damage.
- Examples:**
 - Self-driving Cars**
(failure of the braking system may cause)
 - Emergency Lighting System**
(system failure could lead to injuries during an evacuation)
 - Flight auto-pilot control system
 - Nuclear reactor management system
 - _____

b. Mission-Critical Systems

- A **system failure** may result in the failure of some goal-directed activity.
- Examples:**
 - GPS (location1 $\square$ loc2)**
(Using GPS, the goal is ...)
(If GPS failed...)
 - Traffic control system**
(The goal of TCS is ...)
(If TCS failed ...)
 - Attendance system**
(The goal of attendance_sys is ...)
(If attendance_sys failed ...)

c. Business-Critical Systems

- Systems **failure** can have **serious consequences** for the business
- Examples:**
 - ERP System**
(ERP system **failure** can result in **serious consequences** for the business! agree?)
 - Online banking system**
(OBS **failure** also can have **serious consequences** for the business-Yes/No?)
 - E-commerce website**

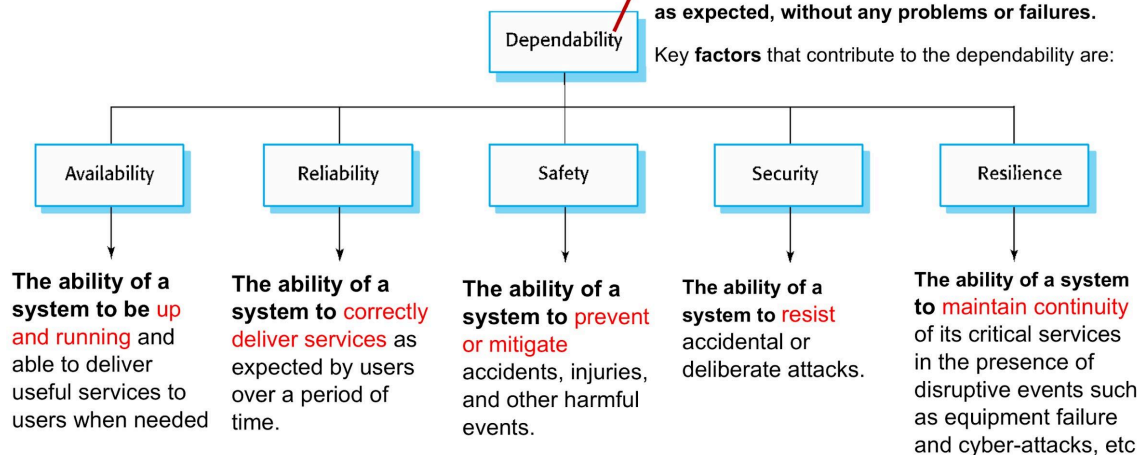


Dimensions of Dependability

8.2 Dependability

Dependability refers to how reliable and consistent a system is. A **dependable system** is one that will work as expected, without any problems or failures.

Key **factors** that contribute to the dependability are:



1. Mission-Critical System - A system failure may result in the failure of some goal-directed activity: Air Traffic Control (ATC) System

ATC systems coordinate aircraft in flight and on the ground to prevent collisions, maintain efficient flight paths, and ensure passenger safety.

- **Most Important Dimension of Dependability: Reliability. (security also can)**
- **Reliability: The ability of a system to correctly deliver services as expected by users over a period of time.**

Any failure in delivering accurate flight path data or communication can lead to mid-air collisions or near misses. The system must consistently deliver correct services over time. Safety is critical too, but reliability underpins it—if the system consistently fails, safety cannot be achieved.

2. Business-Critical System - a system failure can have serious consequences for the business: Online Banking System
Banking systems manage billions in financial transactions daily. Failures can result in economic losses, reputational damage, and legal consequences.

- **Most Important Dimension of Dependability: Availability. (security and reliability also can)**
- **Availability - the ability of a system to be up and running and able to deliver useful services to users when needed.**

Customers expect 24/7 access to transfer funds, pay bills, or withdraw cash. If the system is down, it directly impacts customer trust and the bank's operations. Even a short outage during peak hours can cost millions in lost transactions and litigation

Question 2
Tan Wen Tik

Question 2

Consider the following scenario:

The Food and Drug Administration (FDA) has approved the first automated insulin delivery system. An automated insulin delivery (AID) system has three main components: an insulin pump, a continuous glucose monitor and an algorithm that determines the best insulin dose to keep patients in range.

This represents a step toward a so-called artificial pancreas that could automatically regulate blood sugar levels for people who have diabetes. The device made by the manufacturer Medtronic has been approved to treat people with Type 1 diabetes who are 14 and older.

(Source: <https://www.livescience.com/56305-automated-insulin-delivery-device-approved.html>)

a) Identify ONE (1) important dependability dimension that can be applied to the automated insulin-delivery system. Explain your answer.

b) Insulin delivery system is considered to be which type of *safety critical system*? Explain your answer.

Primary safety-critical systems are those where failure can directly cause hardware failure and threaten people, such as a flight control system. Secondary safety-critical systems are systems whose failure leads to faults in other systems, which then pose a threat to users, such as a sewage treatment system failing and causing contamination. The key difference lies in the directness of the failure and its consequences: **primary failures have immediate, direct impacts, while secondary failures have indirect, cascading effects.**

Primary Safety-Critical Systems

- **Definition:** Systems where a failure directly and immediately results in a hardware failure or a hazardous situation that threatens users.
- **Failure Impact:** The failure is direct and causes the immediate danger.
- **Examples:**
 - An airplane's flight control system, where a failure could directly lead to a crash.
 - A life-support system.
 - An embedded controller in a medical device that directly controls patient care.

Secondary Safety-Critical Systems

Definition: Systems that, when they fail, introduce faults into other systems, which can then have safety consequences for users.

Failure Impact: The failure has an indirect impact, leading to a chain of events that could result in harm.

Examples:

- A sewage treatment system, where a failure could lead to environmental contamination and harm to people.
- A computer-aided design (CAD) system for engineering, where a malfunction could lead to faulty designs being implemented in the real world.
- An electricity distribution system, where a failure might not have an immediate impact but could lead to dangerous situations if

a.

Safety is the most important dimension.

- The system delivers insulin directly to patients. Any incorrect dosage (too high or too low) can cause life-threatening hypoglycemia or hyperglycemia. Therefore, the system must operate in a way that avoids harm, regardless of reliability or availability

b.

It is a Primary Safety-Critical System.

- The failure of the insulin pump directly threatens human life. If the pump malfunctions, the patient could suffer severe complications or death. This is unlike secondary safety-critical systems, which only indirectly cause risks

Question 3 (Leow Yan Luen)

Propose a fault management approach used to manage the faults that might occur in the automated insulin delivery system. Explain your answer with an example.

Fault Management
8.4 Availability & Reliability

3 approaches to improve system reliability:

- Fault avoidance
 - Develop techniques to either minimize the possibilities of mistakes and/or trap mistakes before these result in system failure
 - e.g. **avoid error-prone programming language constructs like pointers, use static analysis to detect program anomalies.**
- Fault detection
 - **Use verification and validation techniques to discover and remove faults in a system before it is deployed.**
 - e.g. implement systematic system testing and debugging.
- Fault tolerance
 - Incorporate **self-checking facilities in a system**
 - **Use of redundant system.**
e.g. check available memory and disk space ...
e.g. run a secondary system at the same time with the primary system.

check program codes to find unusual or problematic patterns.

check whether the software **development process** is carry out correctly

check whether **developed system** is working as expected.



Approach: Fault Detection

- Implement built-in checks to detect errors before they lead to failures.
- Example:

- Include range checks on insulin dosage calculations to ensure they are within safe limits (e.g., not exceeding a maximum dose per hour or per day).
- Use hardware diagnostics (e.g., sensor calibration checks) to detect physical faults.